UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 011 · Lagrange's method of undetermined multipliers · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011·5	Deck 011, slide 5 (rendered .pptx)
T011a 00:01:08	Transcript “Lagrange's method of undetermined coefficient” (theory lecture), segment start
T011b 00:05:03	Transcript “Problems on Lagrange's method of undetermined multipliers”
CW U1 P33	Unit 1 class-work problem 33
Extra Q57	Extra question list “lagrange-ques”, question 57
ESA Jul-23 1c	Past paper: Jul 2023, question 1.c

Contents

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2. 1 · Lagrange's method: the auxiliary function and the procedure
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4. Part B · Symbol glossary

A Formula sheet

1 Lagrange's method: the auxiliary function and the procedure

E1.1 · Auxiliary (Lagrange) function ESA ★★★

$$F(x,y,z) = f(x,y,z) + \lambda\,\phi(x,y,z)$$

GIVES	One function whose stationary points are the constrained stationary points.
APPLIES	Optimise f subject to $\phi(x,y,z)=0$.
NOTE	Write the constraint with everything on one side: $x^2+y^2+z^2-r^2$, $xy+2yz+2zx-432$.

S011·3 · T011a 00:01:08

E1.2 · The four equations ESA ★★★

$$F_x = f_x + \lambda\phi_x = 0, \quad F_y = f_y + \lambda\phi_y = 0, \quad F_z = f_z + \lambda\phi_z = 0, \quad \phi = 0$$

GIVES x, y, z and λ (four unknowns, four equations).

APPLIES Always; with two variables drop F_z (three equations).

NOTE Equivalently $\nabla f = -\lambda\nabla\phi$: the gradients are parallel (added for clarity).

S011·3 · T011a 00:02:17 · T011a 00:02:30

E1.3 · Solving trick: make λ the subject ESA ★★

$$\lambda = -\frac{f_x}{\phi_x} = -\frac{f_y}{\phi_y} = -\frac{f_z}{\phi_z}$$

GIVES Relations between x, y, z without λ.

APPLIES When $\phi_x, \phi_y, \phi_z \neq 0$.

NOTE Or multiply the equations by x, y, z and subtract (slides 011·6, 011·11).

S011·13 · S011·14 · T011b 00:07:24

2 Geometric problems: inscribed solids, nearest and farthest points

E2.1 · Largest box in a sphere and in an ellipsoid ESA ★★★

$$x^2 + y^2 + z^2 = r^2 : \; x = y = z = \tfrac{r}{\sqrt{3}}, \; V_{\max} = \tfrac{8r^3}{3\sqrt{3}}$$
$$\tfrac{x^2}{a^2} + \tfrac{y^2}{b^2} + \tfrac{z^2}{c^2} = 1 : \; x = \tfrac{a}{\sqrt{3}}, \; y = \tfrac{b}{\sqrt{3}}, \; z = \tfrac{c}{\sqrt{3}}, \; V_{\max} = \tfrac{8abc}{3\sqrt{3}}$$

GIVES Half-edges and the largest volume $V = 8xyz$.

APPLIES Box centred at the origin, edges $2x, 2y, 2z$ parallel to the axes.

NOTE The sphere is the ellipsoid with $a = b = c = r$: the box is a cube.

S011·5 · ESA May-24 1c · ESA Jan-24 s3 1a · ESA Jan-24 s4 1a

E2.2 · Distance problems: use the square ESA ★★★

$$f = x^2 + y^2 + z^2 \;\; \text{(origin)}, \qquad f = (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2$$

GIVES The squared distance, which has its extrema at the same points as the distance.

APPLIES Nearest or farthest point on a curve or surface.

NOTE Take the square root at the end.

ADDED FOR CLARITY S011·8 · ESA Jul-23 1c · CW U1 P36

B Symbol glossary

Symbol	Name	Meaning	Appears in
$f(x, y, z)$	Objective	Function to maximise or minimise	E1.1
$\phi(x, y, z)$	Constraint	$\phi = 0$; printed $\varnothing$ on the slides	E1.1, E1.2
λ	Lagrange multiplier	Undetermined constant; not in the answer	E1.1–E1.3
F	Auxiliary function	$f + \lambda\phi$	E1.1
r	Radius	Of the sphere $x^2 + y^2 + z^2 = r^2$; not f_{xx}	E2.1
$a,\ b,\ c$	Semi-axes	Of the ellipsoid (in W3.4: positive constants)	E2.1, W3.4
$2x,\ 2y,\ 2z$	Edges	Of the inscribed box; volume $8xyz$	E2.1

Remark. The slides write the constraint as $\phi(x, y, z) = 0$ (also printed as $\varnothing$) and the auxiliary function as $F = f + \lambda\phi$; the lecture also calls it the Lagrangian. λ is the Lagrange multiplier; its value is found but never needed in the answer. In the sphere problems r is the radius of the sphere, not the second partial f_{xx} of class 009.